LAB-08

Task-1

```
PS D:\Data Structure\Lab Task8> cd "d:\Data St
Preorder: 45 25 15 35 30 65 55 75
Inorder: 15 25 30 35 45 55 65 75
Postorder: 15 30 35 25 55 75 65 45
Nodes with one child: 1
Nodes with two children: 3
Height (edges): 3
PS D:\Data Structure\Lab Task8>
```

Task-2

```
PS D:\Data Structure\Lab Task8> cd "d:\Data St
After deleting 20: 30 35 40 45 50 60 70 80
After deleting 30: 35 40 45 50 60 70 80
After deleting 50: 35 40 45 60 70 80
PS D:\Data Structure\Lab Task8>
```

Task-3

```
PS D:\Data Structure\Lab Task8> cd "d:\Data Structure
IDs in range: 25 30 40 50 60 70
Count: 6
After deleting outside range: 25 30 40 50 60 70
PS D:\Data Structure\Lab Task8>
```

Task-4

```
PS D:\Data Structure\Lab Task8> cd "d:\Data Structure\Lab Task8\
Original inorder: 10 15 25 30 32 33 35 40 45 55 60 65 75 80
Minimum marks: 10
Maximum marks: 80
Threshold: 45
Marks greater than threshold: 55 60 65 75 80
Nodes with two children: 4
Longest root-to-leaf path: 45 25 35 30 32 33
Height (edges): 5
Is tree skewed? No
Number of nodes deleted: 5
Final inorder: 10 25 33 35 40 45 60 65 80
4th smallest remaining mark: 35
PS D:\Data Structure\Lab Task8>
```